

```
// BootRunner.cs
using System.Collections;
using Sy.Core;
using UnityEngine;
using UnityEngine.SceneManagement;
using Logger = Sy.Core.Logger;
namespace Sy.P05.Boot
{
    public static class SceneIds
    {
        public const string Boot = "S00_Boot";
        public const string Main = "S05_Main";
    }
    /// <summary>
    /// S00_Boot      Mock → →      T001④
    ///      Boot      UI T001b  Boot      =
    /// </summary>
    public sealed class BootRunner : MonoBehaviour
    {
        IEnumerator Start()
        {
            BootServices.RegisterAll();
            var sequence = new BootSequence(SwitchToMain);
            if (!sequence.RunAll())
            {
                Logger.Error("BOOT", "boot sequence failed - stay on S00_Boot");
                yield break;
            }
            yield return null;
        }
        bool SwitchToMain()
        {
            StartCoroutine(LoadMainRoutine());
            return true;
        }
        IEnumerator LoadMainRoutine()
        {
            var op = SceneManager.LoadSceneAsync(SceneIds.Main, LoadSceneMode.Single);
            while (op != null && !op.isDone) yield return null;
        }
    }
}

// BootSequence.cs
using System;
using System.Collections.Generic;
using Sy.Core;
using Sy.Platform;
namespace Sy.P05.Boot
{
    /// <summary>
```

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/// Boot      T001
/// ①Logger→② →③Config→④Save→⑤ILocale→⑥Ads.Preload→⑦      [BOOT] n/7
/// ⑦      Play =SceneManager.LoadSceneAsync Editor dry-run=EditorSceneManager.OpenScene
/// X006      SY MOCK
/// &lt;/summary&gt;
public sealed class BootSequence
{
    public const int TotalSteps = 7;
    readonly Func&lt;bool&gt; _switchScene;
    readonly List&lt;string&gt; _lines = new();
    public BootSequence(Func&lt;bool&gt; switchScene)
    {
        _switchScene = switchScene;
    }
    public IReadOnlyList&lt;string&gt; Lines =&gt; _lines;
    public bool RunAll()
    {
        bool ok = true;
        ok &= Run(1, "Logger", LoggerStep);
        ok &= Run(2, "Time", TimeStep);
        ok &= Run(3, "Config", ConfigStep);
        ok &= Run(4, "Save", SaveStep);
        ok &= Run(5, "ILocale", LocaleStep);
        ok &= Run(6, "Ads.Preload", AdsStep);
        ok &= Run(7, "SwitchMain", () =&gt; _switchScene != null && _switchScene());
        return ok;
    }
    bool Run(int n, string name, Func&lt;bool&gt; step)
    {
        bool result;
        try
        {
            result = step();
        }
        catch (Exception e)
        {
            Logger.Error("BOOT", $"step {n} {name} exception: {e.Message}");
            result = false;
        }
        string line = $"[BOOT] {n}/{TotalSteps} {name} {(result ? "ok" : "FAIL")}";
        _lines.Add(line);
        Logger.Info("BOOT", line);
        return result;
    }
    static bool LoggerStep() =&gt; true; //      JS sink platform T001c
    static bool TimeStep()
    {
        var time = ServiceLocator.Get&lt;ITimeService&gt;();
        return time.ServerNowMs() &gt; 0 && !time.TimeCheated();
    }
}

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        static bool ConfigStep()
        {
            var config = ServiceLocator.Get<IConfigService>();
            var update = config.CheckUpdate("mock://manifest");
            return update.NewVersion != null; // mock= X009 manifest
        }
        static bool SaveStep()
        {
            var save = ServiceLocator.Get<ISaveService>();
            save.Save("boot_probe", "{}");
            return !string.IsNullOrEmpty(save.Load("boot_probe"));
        }
        static bool LocaleStep()
        {
            var locale = ServiceLocator.Get<ILocaleService>();
            return !string.IsNullOrEmpty(locale.GetFontSet().NormalFont); // +
        }
        static bool AdsStep()
        {
            ServiceLocator.Get<IAdsService>().Preload("boot_warmup");
            return true;
        }
    }
}

// BootServices.cs
using System;
using System.Collections.Generic;
using System.Threading.Tasks;
using Sy.Core;
using Sy.Platform;
using UnityEngine;
using Logger = Sy.Core.Logger;
namespace Sy.P05.Boot
{
    /// <summary>
    /// Mock SY MOCK X006 Mock X002a AppID touristappid
    /// T001c WX/TT Channel Editor Boot
    /// </summary>
    public static class BootServices
    {
        public static void RegisterAll()
        {
            if (ServiceLocator.TryGet<ITimeService>(out _)) return; //
            ServiceLocator.Register<ITimeService>(new MockTimeService());
            ServiceLocator.Register<IConfigService>(new MockConfigService());
            ServiceLocator.Register<ISaveService>(new MockSaveService());
            ServiceLocator.Register<ILocaleService>(new MockLocaleService());
            ServiceLocator.Register<IAdsService>(new MockAdsService());
            Logger.Info("BOOT", "mock services registered (SY MOCK)");
        }
    }
}

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}
public sealed class MockTimeService : ITimeService
{
    public TimeSource Source => TimeSource.Mock;
    public long ClientNowMs() => DateTimeOffset.UtcNow.ToUnixTimeMilliseconds();
    /// <summary>SY MOCK X006 TCB serverTime→HTTPS Date → T001b </summary>
    public long ServerNowMs() => ClientNowMs();
    public bool TimeCheated() => Math.Abs(ServerNowMs() - ClientNowMs()) > 120_000;
}
public sealed class MockConfigService : IConfigService
{
    public T GetTable<T>(string tableId) where T : class
        => throw new KeyNotFoundException("mock config table not loaded: " + tableId);
    public bool TryGetTable<T>(string tableId, out T table) where T : class
    {
        table = null;
        return false;
    }
    public ConfigUpdateResult CheckUpdate(string manifestUrl)
        => new ConfigUpdateResult(false, "mock-0.0.1");
    public void SwapPolicy(ConfigSwapPolicy policy) { }
    public string PinConfigVersion(string scope) => "mock-0.0.1";
}
public sealed class MockSaveService : ISaveService
{
    const string Prefix = "sy_save_";
    public void Save(string slot, string payloadJson) => PlayerPrefs.SetString(Prefix + slot, payloadJson);
    public string Load(string slot) => PlayerPrefs.GetString(Prefix + slot, string.Empty);
    /// <summary>X006 T001b v1→v2 </summary>
    public SyncResult CloudSync() => new SyncResult(false, null);
}
public sealed class MockLocaleService : ILocaleService
{
    public void SetLang(string lang) { }
    /// <summary>key key T001b cfg_i18n + NUnit </summary>
    public string GetText(string key, params string[] args)
    {
        Logger.Warn("LOCALE", "missing i18n key (fallback): " + key);
        return key;
    }
    public FontSet GetFontSet()
        => new FontSet("LegacyRuntime.ttf", "LegacyRuntime.ttf", "LegacyRuntime.ttf");
}
public sealed class MockAdsService : IAdsService
{
    public event Action<AdLifecycle> OnAdEvent;
    public Task<RewardResult> ShowRewarded(string pointId)
        => Task.FromResult(new RewardResult { Completed = true, Reason = "mock" });
    public void ShowInterstitial(string pointId) { }
    public void ShowBanner(string pointId) { }
}

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        public void Preload(string pointId)
        {
            Logger.Info("ADS", "preload (mock): " + pointId);
            OnAdEvent?.Invoke(new AdLifecycle(AdLifecycleType.Fill, pointId, "mock"));
        }
    }
}

// MainUI.cs
using UnityEngine;
using UnityEngine.SceneManagement;
namespace Sy.P05.Newbie
{
    public static class NewbieSceneIds
    {
        public const string Main = "S05_Main";
        public const string Battle = "S05_Battle";
    }
    /// <summary>S05          →S05_Battle M5          —          </summary>
    public sealed class MainUI : MonoBehaviour
    {
        public void StartSurvival()
        {
            SceneManager.LoadScene(NewbieSceneIds.Battle);
        }
        void Awake()
        {
            Application.targetFrameRate = 30;
            Sy.Core.MainArtInjector.Apply(transform, "P05_scene_01", "P05_general_01_zhaoyun_awake");
        }
    }
}

// BattleLauncher.cs
using System.Collections.Generic;
using UnityEngine;
namespace Sy.P05.Newbie
{
    /// <summary>
    /// S05_Battle          FurnaceSim+TutorialFlow+FurnaceView          Tick
    /// Sy.Telemetry Editor =Debug sink T001e          +TCB
    /// HUD          6
    /// </summary>
    public sealed class BattleLauncher : MonoBehaviour
    {
        FurnaceSim _sim;
        TutorialFlow _tutorial;
        FurnaceView _view;
        Sy.Telemetry.TelemetryService _telemetry;
        int _lastCoins = -1, _lastCitizens = -1, _lastWood = -1, _lastFuel = -1;
        void Awake()
        {

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Application.targetFrameRate = 30;
var canvas = FindObjectOfType<Canvas>();
_sim = new FurnaceSim();
_view = gameObject.AddComponent<FurnaceView>();
_view.Initialize(canvas, _sim);
_telemetry = new Sy.Telemetry.TelemetryService();
_telemetry.AddSink(new TelemetryDebugSink());
_tutorial = new TutorialFlow(
    _sim,
    hint => {
        Debug.Log("[HINT] " + hint);
        _view.SetHint(hint);
    },
    step => { _telemetry.Track(Sy.Telemetry.EventIds.NewbieStep,
        new Dictionary<string, string> { { "step", step.ToString() } } });
    _view.ClaimRequested += OnClaimRequested;
    _view.FirewoodRequested += OnFirewoodRequested;
    _view.BuildRequested += OnBuildRequested;
    _view.DigRequested += OnDigRequested;
    _view.NearDispatchRequested += () => OnDispatchRequested(0);
    _view.FarDispatchRequested += () => OnDispatchRequested(1);
    _view.SocialTapped += OnSocialTapped;
    _sim.PhaseChanged += OnPhaseChanged;
    _sim.DigStageReached += OnDigStageReached;
}
void Update()
{
    _sim.Tick(Time.deltaTime);
    _view.SyncView(_sim);
    if (_sim.Coins != _lastCoins) { _lastCoins = _sim.Coins; _view.SetCoins(_lastCoins); }
    if (_sim.Citizens != _lastCitizens) { _lastCitizens = _sim.Citizens; _view.SetCitizens(_lastCitizens); }
    if (_sim.Wood != _lastWood) { _lastWood = _sim.Wood; _view.SetWood(_lastWood); }
    if ((int)_sim.Fuel != _lastFuel) { _lastFuel = (int)_sim.Fuel; _view.SetFuel(_lastFuel); }
}
void OnClaimRequested()
{
    _sim.ClaimOffline();
}
void OnFirewoodRequested()
{
    _sim.AddFirewood();
}
void OnBuildRequested()
{
    _sim.BuildWoodyard();
}
void OnDigRequested()
{
    _sim.StartDig();
}
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    }
    void OnDispatchRequested(int node)
    {
        _sim.DispatchOrder(node);
    }
    void OnSocialTapped()
    {
        _view.SetHint("          —          1 / .          ");
    }
    void OnDigStageReached(int stage)
    {
        _view.SetDigStage(stage);
    }
    void OnPhaseChanged(FurnaceSim.SimPhase phase)
    {
        if (phase == FurnaceSim.SimPhase.Settled)
            _view.ShowSettle(_sim.Coins, _sim.Citizens, _sim.Wood, (int)_sim.Fuel, _sim.Day, _sim.OutsideDispatchDon
        else
            _view.HideIntro();
    }
    sealed class TelemetryDebugSink : Sy.Telemetry.TelemetryService.ISink
    {
        public void Write(string evtId, Dictionary<string, string> fields)
        {
            var sb = new System.Text.StringBuilder(evtId);
            foreach (var kv in fields)
            {
                sb.Append(' ').Append(kv.Key).Append('=').Append(kv.Value);
            }
            Debug.Log("[TELEMETRY] " + sb);
        }
    }
}

// FurnaceSim.cs
using System;
namespace Sy.P05.Newbie
{
    /// <summary>
    ///          C#          Play  Editor headless  NewbieFlowCheck
    ///          §4          ticking (fuelPerMin=4+fireLv*2·          )/  lv1  /  →  /
    ///          (  x2+frostRisk+          )/          /
    ///          06 05  M0          →M1  (8h )+  ( 1 · +30)→M2  lv1
    /// (  +  lv2·          )→M3  (10s )  →M4  +  (  /          )
    /// →M5  +  +  "  "(  +1)
    ///          1          =
    /// </summary>
    public sealed class FurnaceSim
    {
        public enum SimPhase

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{
    Intro, AwaitClaim, AwaitFirewood, AwaitBuild, AwaitDig, Digging,
    AwaitDispatch, Dispatching, Reveal, Settled
}

public SimPhase Phase { get; private set; } = SimPhase.Intro;
public float Fuel { get; private set; } = NewbieConfig.StartFuel;
public int Coins { get; private set; }
public int Citizens { get; private set; } = NewbieConfig.StartCitizens;
public int Wood { get; private set; } = NewbieConfig.StartWood;
public int Day { get; private set; } = NewbieConfig.StartDay;
public int FireLevel { get; private set; } = 1; // §4.2① →
public bool HasWoodyard { get; private set; }
public bool SurvivorFound { get; private set; }
public float PhaseTimer { get; private set; }
public float LightRadius => NewbieConfig.LightRadiusByLevel[FireLevel];
public int DigStagesSeen { get; private set; } // bit0-3=
public bool OutsideDispatchDone { get; private set; } // ×2+ +
public int AdPopupCount => 0; //
public event Action<SimPhase> PhaseChanged;
public event Action Claimed;
public event Action FirewoodAdded;
public event Action WoodyardBuilt;
public event Action FireLevelUp; // =M5
public event Action DigStarted;
public event Action<int> DigStageReached; // V1.2 ②
public event Action SurvivorDug;
public event Action<int> DispatchStarted; // ( 0 /1 )
public event Action<int, int> OrderDone; // ( , )
public event Action FrostRiskWarned; // frostRisk
public event Action Settled;
int _dispatchNode;
int _digStageDone;
float _woodTimer;
public void Tick(float dt)
{
    if (Phase == SimPhase.Intro)
    {
        PhaseTimer += dt;
        if (PhaseTimer >= NewbieConfig.IntroSeconds) SetPhase(SimPhase.AwaitClaim);
        return;
    }
    if (Phase == SimPhase.Settled) return;
    // ticking §4.1 fuelPerMin=4+fireLv*2 1
    Fuel = Math.Max(NewbieConfig.FuelFloor, Fuel - NewbieConfig.FuelDrainPerSecond(FireLevel) * dt);
    // lv1 WoodyardWoodInterval +1 WoodyardWoodCap
    if (HasWoodyard)
    {
        _woodTimer += dt;
        if (_woodTimer >= NewbieConfig.WoodyardWoodInterval && Wood < NewbieConfig.WoodyardWoodCap)
        {

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        _woodTimer = 0f;
        Wood++;
    }
}
switch (Phase)
{
    case SimPhase.Digging:
        PhaseTimer += dt;
        TickDigStages();
        if (PhaseTimer >= NewbieConfig.DigSeconds)
        {
            SurvivorFound = true;
            Citizens++;
            SurvivorDug?.Invoke();
            SetPhase(SimPhase.AwaitDispatch);
        }
        break;
    case SimPhase.Dispatching:
        PhaseTimer += dt;
        if (PhaseTimer >= NewbieConfig.DispatchSeconds)
        {
            Wood += NewbieConfig.DispatchYield[_dispatchNode];
            if (_dispatchNode == 1)
            {
                OutsideDispatchDone = true;
                Fuel = Math.Max(NewbieConfig.FuelFloor, Fuel - NewbieConfig.OutsideWarmupFuel); //
            }
            OrderDone?.Invoke(NewbieConfig.DispatchYield[_dispatchNode], _dispatchNode);
            SetPhase(SimPhase.Reveal);
        }
        break;
    case SimPhase.Reveal:
        PhaseTimer += dt;
        if (PhaseTimer >= NewbieConfig.RevealSeconds)
        {
            Day++; // → +1 §4.1 +
            Settled?.Invoke();
            SetPhase(SimPhase.Settled);
        }
        break;
}
}
void TickDigStages()
{
    int stage = Math.Min(NewbieConfig.DigStages.Length,
        1 + (int)(PhaseTimer / (NewbieConfig.DigSeconds / NewbieConfig.DigStages.Length)));
    if (stage > _digStageDone)
    {
        _digStageDone = stage;
        DigStagesSeen |= 1 && (stage - 1);
    }
}

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        DigStageReached?.Invoke(stage);
    }
}
/// <summary>M1a      8h      </summary>
public bool ClaimOffline()
{
    if (Phase != SimPhase.AwaitClaim) return false;
    Coins += NewbieConfig.OfflineCoins;
    Claimed?.Invoke();
    SetPhase(SimPhase.AwaitFirewood);
    return true;
}
/// <summary>M1b      1 → +30      </summary>
public bool AddFirewood()
{
    if (Phase != SimPhase.AwaitFirewood) return false;
    if (Wood < 1) return false; //      3 —
    Wood--;
    Fuel = Math.Min(100f, Fuel + NewbieConfig.FirewoodFuel);
    FirewoodAdded?.Invoke();
    SetPhase(SimPhase.AwaitBuild);
    return true;
}
/// <summary>M2      lv1      + +      lv2      ·      ① </summary>
public bool BuildWoodyard()
{
    if (Phase != SimPhase.AwaitBuild) return false;
    HasWoodyard = true;
    FireLevel++;
    FireLevelUp?.Invoke();
    WoodyardBuilt?.Invoke();
    SetPhase(SimPhase.AwaitDig);
    return true;
}
/// <summary>M3      →10s      </summary>
public bool StartDig()
{
    if (Phase != SimPhase.AwaitDig) return false;
    PhaseTimer = 0f;
    _digStageDone = 0;
    DigStarted?.Invoke();
    SetPhase(SimPhase.Digging);
    return true;
}
/// <summary>M4      0=      /1=      · x2·frostRisk+      </summary>
public bool DispatchOrder(int node)
{
    if (Phase != SimPhase.AwaitDispatch) return false;
    if (node < 0 || node >= NewbieConfig.DispatchNames.Length) return false;
    _dispatchNode = node;
}

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        if (node == 1)
        {
            FrostRiskWarned?.Invoke(); // frostRisk=0.05* 15% —
        }
        PhaseTimer = 0f;
        DispatchStarted?.Invoke(node);
        SetPhase(SimPhase.Dispatching);
        return true;
    }
    void SetPhase(SimPhase p)
    {
        if (Phase == p) return;
        Phase = p;
        PhaseTimer = 0f;
        PhaseChanged?.Invoke(p);
    }
}

// FurnaceView.cs
using System;
using System.Collections;
using UnityEngine;
using UnityEngine.UI;
namespace Sy.P05.Newbie
{
    /// <summary>
    ///     UGUI— x x x x
    ///     UI Initialize      YAML      = / /      → · ① /
    ///     vs                /      / " "
    ///     =
    /// </summary>
    public sealed class FurnaceView : MonoBehaviour
    {
        Canvas _canvas;
        Sprite _square;
        FurnaceSim _sim;
        RectTransform _lightCircle, _furnaceRt;
        Image _fuelFill, _digFill, _dispatchFill;
        Image _digPortrait; // X027: dig-stage general art (frozen peek -&gt; awake reveal)
        Button _firewoodButton, _buildButton, _iceButton, _nearButton, _farButton, _socialButton;
        Image _firewoodImage, _buildImage, _iceImage, _nearImage, _farImage;
        Text _coinText, _citizenText, _woodText, _fuelText, _hintText, _digStageText;
        RectTransform _offlinePopup, _settlePanel;
        Text _settleStats;
        FurnaceSim.SimPhase _lastPhase;
        float _circleScale = NewbieConfig.LightRadiusByLevel[1];
        public event Action ClaimRequested;
        public event Action FirewoodRequested;
        public event Action BuildRequested;
        public event Action DigRequested;
    }
}

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public event Action NearDispatchRequested; // .
public event Action FarDispatchRequested; // . x2.
public event Action SocialTapped;
static readonly Color SnowWhite = new Color(0.969f, 0.976f, 0.988f); // #F7F9FC
static readonly Color IceBlue = new Color(0.659f, 0.773f, 0.855f); // #A8C5DA
static readonly Color InkBlue = new Color(0.118f, 0.227f, 0.373f); // #1E3A5F
static readonly Color FurnaceOrange = new Color(1f, 0.42f, 0.102f); // #FF6B1A
static readonly Color IceGray = new Color(0.42f, 0.48f, 0.549f); // #6B7B8C
public void Initialize(Canvas canvas, FurnaceSim sim)
{
    _canvas = canvas;
    _sim = sim;
    _square = MakeSquareSprite();
    BuildScene();
    BuildHud();
    BuildOfflinePopup();
    BuildSettlePanel();
    SetHint(".....");
    StartCoroutine(SnowRoutine());
    StartCoroutine(FlameRoutine());
}
// ——
void BuildScene()
{
    var bg = MakeFullImage("Bg", InkBlue);
    bg.SetAsFirstSibling();
    // X027 generated art: swap the solid Bg block for the scene painting
    // when present; missing art keeps the procedural look untouched.
    var bgImg = bg.GetComponent<Image>();
    var bgArt = Sy.Core.ArtRuntime.Get("P05_scene_01");
    if (bgArt != null)
    {
        bgImg.sprite = bgArt;
        bgImg.color = Color.white;
        bgImg.preserveAspect = true;
        bgImg.raycastTarget = false;
    }
    //
    var ground = MakeCanvasChild("Ground");
    ground.sizeDelta = new Vector2(1080, 520);
    ground.anchoredPosition = new Vector2(0, -700);
    var groundImg = ground.GetComponent<Image>();
    groundImg.sprite = _square;
    groundImg.color = new Color(0.85f, 0.88f, 0.92f);
    // → lv1 0.55 → lv2 0.95
    _lightCircle = MakeCanvasChild("LightCircle");
    _lightCircle.sizeDelta = new Vector2(700, 700);
    _lightCircle.anchoredPosition = new Vector2(0, -420);
    var circleImg = _lightCircle.GetComponent<Image>();
    circleImg.sprite = _square;

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circleImg.color = new Color(IceBlue.r, IceBlue.g, IceBlue.b, 0.10f);
// X027 generated art: hidden general portrait shown by dig stage below.
var portraitGo = new GameObject("DigPortrait", typeof(RectTransform), typeof(Image));
portraitGo.transform.SetParent(_canvas.transform, false);
_digPortrait = portraitGo.GetComponent<Image>();
var portraitRt = (RectTransform)portraitGo.transform;
portraitRt.anchorMin = portraitRt.anchorMax = new Vector2(0.5f, 0.5f);
portraitRt.anchoredPosition = new Vector2(0, 40);
portraitRt.sizeDelta = new Vector2(640, 860);
_digPortrait.raycastTarget = false;
_digPortrait.gameObject.SetActive(false);
//
_furnaceRt = MakeCanvasChild("Furnace");
_furnaceRt.anchoredPosition = new Vector2(0, -420);
_furnaceRt.sizeDelta = new Vector2(150, 150);
var furnaceImg = _furnaceRt.GetComponent<Image>();
furnaceImg.sprite = _square;
furnaceImg.color = FurnaceOrange;
MakeText(_furnaceRt, " ", 60, Vector2.zero, TextAnchor.MiddleCenter, 150, 150).color = InkBlue;
var fuelBar = MakeCanvasChild("FuelBar");
fuelBar.anchoredPosition = new Vector2(0, -560);
fuelBar.sizeDelta = new Vector2(300, 24);
var fuelBg = fuelBar.GetComponent<Image>();
fuelBg.sprite = _square;
fuelBg.color = new Color(0.15f, 0.18f, 0.22f);
var fillGo = new GameObject("Fill", typeof(RectTransform), typeof(Image));
fillGo.transform.SetParent(fuelBar, false);
var fillRt = (RectTransform)fillGo.transform;
fillRt.anchorMin = new Vector2(0f, 0.5f);
fillRt.anchorMax = new Vector2(1f, 0.5f);
fillRt.anchoredPosition = Vector2.zero;
fillRt.sizeDelta = new Vector2(0, 24);
_fuelFill = fillGo.GetComponent<Image>();
_fuelFill.sprite = _square;
_fuelFill.type = Image.Type.Filled;
_fuelFill.fillMethod = Image.FillMethod.Horizontal;
_fuelFill.color = FurnaceOrange;
//
_firewoodButton = MakeActionButton("FirewoodButton", " ", new Vector2(-260, -420), FirewoodRequested, out _
//
_buildButton = MakeActionButton("BuildButton", " \n ", new Vector2(300, -420), BuildRequested, out _buildIm
//
_iceButton = MakeActionButton("IceButton", " \n ", new Vector2(-300, -660), DigRequested, out _iceImage);
//      M4      /      . x2.  —
_nearButton = MakeActionButton("NearNode", " . \n+6 ", new Vector2(170, -470), NearDispatchRequested, ou
_farButton = MakeActionButton("FarNode", " . \n+12 15%", new Vector2(450, -260), FarDispatchRequested, o
// X027 generated art: furnace + building paintings on the action blocks.
ApplyActionArt(_furnaceRt.GetComponent<Image>(), "P05_icon_fire");
ApplyActionArt(_firewoodImage, "P05_icon_wood");
ApplyActionArt(_buildImage, "P05_building_01_wood");

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ApplyActionArt(_iceImage, "P05_building_02_ice");
ApplyActionArt(_nearImage, "P05_building_03_farm");
ApplyActionArt(_farImage, "P05_building_04_iron");
//
var digBar = MakeCanvasChild("DigBar");
digBar.anchoredPosition = new Vector2(-300, -540);
digBar.sizeDelta = new Vector2(220, 18);
var digBg = digBar.GetComponent<Image>();
digBg.sprite = _square;
digBg.color = new Color(0.15f, 0.18f, 0.22f);
var digFillGo = new GameObject("Fill", typeof(RectTransform), typeof(Image));
digFillGo.transform.SetParent(digBar, false);
var digFillRt = (RectTransform)digFillGo.transform;
digFillRt.anchorMin = new Vector2(0f, 0.5f);
digFillRt.anchorMax = new Vector2(1f, 0.5f);
digFillRt.anchoredPosition = Vector2.zero;
digFillRt.sizeDelta = new Vector2(0, 18);
_digFill = digFillGo.GetComponent<Image>();
_digFill.sprite = _square;
_digFill.type = Image.Type.Filled;
_digFill.fillMethod = Image.FillMethod.Horizontal;
_digFill.color = IceBlue;
digBar.gameObject.SetActive(false);
//      "      "
_digStageText = MakeText(_canvas.transform, "", 34, new Vector2(-300, -450), TextAnchor.MiddleCenter, 340, 6
_digStageText.color = new Color(0.05f, 0.09f, 0.15f); // U184: deep-ink - was IceBlue-on-snow 1.4:1 (17 do
_digStageText.gameObject.SetActive(false);
//
var dispatchBar = MakeCanvasChild("DispatchBar");
dispatchBar.anchoredPosition = new Vector2(310, -100);
dispatchBar.sizeDelta = new Vector2(220, 18);
var dispBg = dispatchBar.GetComponent<Image>();
dispBg.sprite = _square;
dispBg.color = new Color(0.15f, 0.18f, 0.22f);
var dispFillGo = new GameObject("Fill", typeof(RectTransform), typeof(Image));
dispFillGo.transform.SetParent(dispatchBar, false);
var dispFillRt = (RectTransform)dispFillGo.transform;
dispFillRt.anchorMin = new Vector2(0f, 0.5f);
dispFillRt.anchorMax = new Vector2(1f, 0.5f);
dispFillRt.anchoredPosition = Vector2.zero;
dispFillRt.sizeDelta = new Vector2(0, 18);
_dispatchFill = dispFillGo.GetComponent<Image>();
_dispatchFill.sprite = _square;
_dispatchFill.type = Image.Type.Filled;
_dispatchFill.fillMethod = Image.FillMethod.Horizontal;
_dispatchFill.color = IceBlue;
dispatchBar.gameObject.SetActive(false);
//      M5
_socialButton = MakeActionButton("SocialButton", "      \n      ", new Vector2(380, 300), SocialTapped, out _social
_socialButton.gameObject.SetActive(false);

```

```

}
Image _socialImage;
void BuildHud()
{
    var root = _canvas.transform;
    _coinText = MakeText(root, "  0", 46, new Vector2(0, 880), TextAnchor.UpperLeft, 360, 70);
    _coinText.color = SnowWhite;
    _fuelText = MakeText(root, "  20%", 46, new Vector2(0, 880), TextAnchor.UpperCenter, 380, 70);
    _fuelText.color = FurnaceOrange;
    _citizenText = MakeText(root, "  1", 44, new Vector2(0, 820), TextAnchor.UpperRight, 320, 60);
    _citizenText.color = SnowWhite;
    _woodText = MakeText(root, "  3", 44, new Vector2(0, 760), TextAnchor.UpperRight, 320, 60);
    _woodText.color = SnowWhite; // U184: was IceGray-on-deep ~2.6:1 (17 doc section 3.7)
    _hintText = MakeText(root, "", 40, new Vector2(0, -840), TextAnchor.MiddleCenter, 1020, 150);
    _hintText.color = new Color(0.05f, 0.09f, 0.15f); // U184: deep-ink on snow band - was SnowWhite-on-snow 1
}
void BuildOfflinePopup()
{
    _offlinePopup = MakeOverlayPanel("OfflinePopup", new Color(0.06f, 0.10f, 0.16f, 0.94f));
    MakeText(_offlinePopup, "  8  ", 66, new Vector2(0, 220), TextAnchor.MiddleCenter, 900, 100).color = SnowW
    MakeText(_offlinePopup, "  x240  ", 52, new Vector2(0, 80), TextAnchor.MiddleCenter, 900, 90).color = Ice
    MakePanelButton(_offlinePopup, "  ", new Vector2(0, -160), () => ClaimRequested?.Invoke());
}
void BuildSettlePanel()
{
    _settlePanel = MakeOverlayPanel("SettlePanel", new Color(0.06f, 0.10f, 0.16f, 0.96f));
    MakeText(_settlePanel, "  ", 96, new Vector2(0, 560), TextAnchor.MiddleCenter, 1040, 120).color = Furnac
    _settleStats = MakeText(_settlePanel, "", 44, new Vector2(0, 360), TextAnchor.MiddleCenter, 1020, 380);
    MakePanelButton(_settlePanel, "  ", new Vector2(-260, -300), () => UnityEngine.SceneManagement.SceneMan
    MakePanelButton(_settlePanel, "  ", new Vector2(260, -300), () => UnityEngine.SceneManagement.SceneMan
    _settlePanel.gameObject.SetActive(false);
}
// _____
public void SyncView(FurnaceSim sim)
{
    var phase = sim.Phase;
    if (phase != _lastPhase)
    {
        _lastPhase = phase;
        OnPhaseVisual(phase);
    }
    float fuel01 = sim.Fuel / 100f;
    _fuelFill.fillAmount = fuel01;
    //      →      ① +
    float target = NewbieConfig.LightRadiusByLevel[sim.FireLevel] + fuel01 * 0.05f;
    _circleScale = Mathf.Lerp(_circleScale, target, 0.08f);
    _lightCircle.localScale = Vector3.one * _circleScale;
    if (phase == FurnaceSim.SimPhase.Digging)
        _digFill.fillAmount = sim.PhaseTimer / NewbieConfig.DigSeconds;
    if (phase == FurnaceSim.SimPhase.Dispatching)

```

```

        _dispatchFill.fillAmount = sim.PhaseTimer / NewbieConfig.DispatchSeconds;
        SetButtonState(_firewoodButton, _firewoodImage, phase == FurnaceSim.SimPhase.AwaitFirewood);
        SetButtonState(_buildButton, _buildImage, phase == FurnaceSim.SimPhase.AwaitBuild);
        SetButtonState(_iceButton, _iceImage, phase == FurnaceSim.SimPhase.AwaitDig);
        bool dispatchable = phase == FurnaceSim.SimPhase.AwaitDispatch;
        SetButtonState(_nearButton, _nearImage, dispatchable);
        SetButtonState(_farButton, _farImage, dispatchable);
    }
    void OnPhaseVisual(FurnaceSim.SimPhase phase)
    {
        switch (phase)
        {
            case FurnaceSim.SimPhase.AwaitClaim:
                _offlinePopup.gameObject.SetActive(true);
                break;
            case FurnaceSim.SimPhase.AwaitFirewood:
                _offlinePopup.gameObject.SetActive(false);
                break;
            case FurnaceSim.SimPhase.Digging:
                _digFill.transform.parent.gameObject.SetActive(true);
                _digStageText.gameObject.SetActive(true);
                break;
            case FurnaceSim.SimPhase.AwaitDispatch:
                _digFill.transform.parent.gameObject.SetActive(false);
                _digStageText.gameObject.SetActive(false);
                break;
            case FurnaceSim.SimPhase.Dispatching:
                _dispatchFill.transform.parent.gameObject.SetActive(true);
                break;
            case FurnaceSim.SimPhase.Reveal:
                _dispatchFill.transform.parent.gameObject.SetActive(false);
                _socialButton.gameObject.SetActive(true);
                break;
            case FurnaceSim.SimPhase.Settled:
                _socialButton.gameObject.SetActive(true);
                break;
        }
    }
}

static void SetButtonState(Button button, Image image, bool active)
{
    if (button.interactable == active) return;
    button.interactable = active;
    image.color = active ? FurnaceOrange : new Color(0.30f, 0.34f, 0.40f);
}

// —— HUD ——

public void SetCoins(int c) => _coinText.text = "    " + c;
public void SetCitizens(int n) => _citizenText.text = "    " + n;
public void SetWood(int w) => _woodText.text = "    " + w;
public void SetFuel(int f) => _fuelText.text = "    " + f + "%";
public void SetHint(string hint) => _hintText.text = hint;

```

```

/// <summary>          → → → </summary>
public void SetDigStage(int stage)
{
    _digStageText.text = NewbieConfig.DigStages[stage - 1] + " " + stage * 25 + "% ";
    // X027: frozen silhouette peek at "reveal", full-color awake at "awake".
    if (_digPortrait == null) return;
    var art = stage >= 4 ? Sy.Core.ArtRuntime.Get("P05_general_02_guanyu_awake")
        : stage == 3 ? Sy.Core.ArtRuntime.Get("P05_general_02_guanyu_frozen")
        : null;
    if (art != null)
    {
        _digPortrait.sprite = art;
        _digPortrait.color = stage >= 4 ? Color.white : new Color(1f, 1f, 1f, 0.6f);
        _digPortrait.preserveAspect = true;
        _digPortrait.gameObject.SetActive(true);
    }
    else
    {
        _digPortrait.gameObject.SetActive(false);
    }
}
public void HideIntro()
{
    // M0      Hint
}
public void ShowSettle(int coins, int citizens, int wood, int fuel, int day, bool outside)
{
    _settlePanel.gameObject.SetActive(true);
    _settleStats.text = $" {coins} · {citizens} · {wood} · {fuel}%\n {day} {NewbieConfig.Miles}
}
// ——
/// <summary>X027: swap a solid action block for generated art when the key exists.</summary>
static void ApplyActionArt(Image image, string key)
{
    var art = Sy.Core.ArtRuntime.Get(key);
    if (image == null || art == null) return;
    image.sprite = art;
    image.color = Color.white;
    image.preserveAspect = true;
}
IEnumerator SnowRoutine()
{
    var flakes = new RectTransform[7];
    for (int i = 0; i < flakes.Length; i++)
    {
        var go = new GameObject("Snow_" + i, typeof(RectTransform), typeof(Image));
        go.transform.SetParent(_canvas.transform, false);
        flakes[i] = (RectTransform)go.transform;
        flakes[i].anchorMin = flakes[i].anchorMax = new Vector2(0.5f, 0.5f);
        flakes[i].sizeDelta = new Vector2(18, 18);
    }
}

```

```
        flakes[i].anchoredPosition = new Vector2(UnityEngine.Random.Range(-520f, 520f), UnityEngine.Random.Range(-520f, 520f));
        var img = go.GetComponent<Image>();
        img.sprite = _square;
        img.color = new Color(SnowWhite.r, SnowWhite.g, SnowWhite.b, 0.75f);
        flakes[i].SetAsFirstSibling();
        flakes[i].SetSiblingIndex(1);
    }
    while (true)
    {
        for (int i = 0; i < flakes.Length; i++)
        {
            var pos = flakes[i].anchoredPosition;
            pos.y -= 60f * Time.deltaTime;
            pos.x += Mathf.Sin(Time.time + i) * 10f * Time.deltaTime;
            if (pos.y < -950f)
            {
                pos.y = 950f;
                pos.x = UnityEngine.Random.Range(-520f, 520f);
            }
            flakes[i].anchoredPosition = pos;
        }
        yield return null;
    }
}

IEnumerator FlameRoutine()
{
    while (true)
    {
        float fuel01 = _sim.Fuel / 100f;
        float pulse = 0.5f + 0.5f * Mathf.Sin(Time.unscaledTime * (2f + fuel01 * 3f));
        _furnaceRt.localScale = Vector3.one * (1f + 0.05f + 0.06f * pulse * fuel01);
        yield return null;
    }
}

// _____

RectTransform MakeCanvasChild(string name)
{
    var go = new GameObject(name, typeof(RectTransform), typeof(Image));
    go.transform.SetParent(_canvas.transform, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    return rt;
}

RectTransform MakeFullImage(string name, Color color)
{
    var rt = MakeCanvasChild(name);
    rt.anchorMin = Vector2.zero;
    rt.anchorMax = Vector2.one;
    rt.anchoredPosition = Vector2.zero;
    rt.sizeDelta = Vector2.zero;
}
```

```
        rt.GetComponent<Image>().color = color;
        return rt;
    }
    Button MakeActionButton(string name, string label, Vector2 pos, Action onClick, out Image image)
    {
        var go = new GameObject(name, typeof(RectTransform), typeof(Image), typeof(Button));
        go.transform.SetParent(_canvas.transform, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = new Vector2(160, 160);
        image = go.GetComponent<Image>();
        image.sprite = _square;
        image.color = new Color(0.30f, 0.34f, 0.40f);
        var label1 = MakeText(go.transform, label, 34, Vector2.zero, TextAnchor.MiddleCenter, 170, 170);
        label1.color = SnowWhite;
        var button = go.GetComponent<Button>();
        button.interactable = false;
        button.onClick.AddListener(() => onClick());
        return button;
    }
    Button MakePanelButton(RectTransform parent, string label, Vector2 pos, Action onClick)
    {
        var go = new GameObject(label, typeof(RectTransform), typeof(Image), typeof(Button));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = new Vector2(420, 120);
        var image = go.GetComponent<Image>();
        image.sprite = _square;
        image.color = FurnaceOrange;
        MakeText(go.transform, label, 48, Vector2.zero, TextAnchor.MiddleCenter, 420, 120).color = InkBlue;
        var button = go.GetComponent<Button>();
        button.onClick.AddListener(() => onClick());
        return button;
    }
    RectTransform MakeOverlayPanel(string name, Color color)
    {
        var go = new GameObject(name, typeof(RectTransform), typeof(Image));
        go.transform.SetParent(_canvas.transform, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = Vector2.zero;
        rt.anchorMax = Vector2.one;
        rt.offsetMin = rt.offsetMax = Vector2.zero;
        var image = go.GetComponent<Image>();
        image.sprite = _square;
        image.color = color;
        return rt;
    }
}
```

```

static Text MakeText(Transform parent, string text, int size, Vector2 pos, TextAnchor anchor, float w, float h)
{
    var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = new Vector2(w, h);
    var t = go.GetComponent<Text>();
    t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
    t.text = text;
    t.fontSize = size;
    t.alignment = anchor;
    t.color = new Color(0.91f, 0.89f, 0.85f);
    t.supportRichText = true;
    return t;
}

static Sprite MakeSquareSprite()
{
    var tex = new Texture2D(8, 8, TextureFormat.RGBA32, false);
    var px = tex.GetPixels32();
    for (int i = 0; i < px.Length; i++) px[i] = new Color32(255, 255, 255, 255);
    tex.SetPixels32(px);
    tex.Apply(false, false);
    return Sprite.Create(tex, new Rect(0, 0, 8, 8), new Vector2(0.5f, 0.5f), 8f);
}
}

// NewbieConfig.cs
namespace Sy.P05.Newbie
{
    /// <summary>
    /// P05 . V0.2 — X009+IConfigService Design/configs/P05 JSON
    /// V1.3§4.1 ticking (fuelPerMin="4+fireLv*2". )/§4.2 ① = ( → /
    /// x2.frostRisk="0.05*distFromFire")/② ( → → → —"4h " " " )/
    /// D1-D2 / + 06 05 M0-M5
    /// #F7F9FC x #A8C5DA x #1E3A5F x #FF6B1A x #6B7B8C
    /// 10 →10 →10s §4.2
    /// </summary>
    public static class NewbieConfig
    {
        public const string GameTitle = " ";
        public const string GameSlogan = " ";
        public const float IntroSeconds = 3f; // M0 +
        public const float DigSeconds = 10f; // M3 10 .
        public const float DispatchSeconds = 10f; // M4
        public const float RevealSeconds = 2f; // M5
        public const float MaxTutorialSeconds = 210f; // ≤3.5min 06 ②
        // — §4.1 fuelPerMin="4+fireLv*2" —
        public const float StartFuel = 20f; // M0
    }
}

```

```

    public const float FuelFloor = 1f; //
    public const float FirewoodFuel = 30f; // M1 +30 1
    public const int StartWood = 3; //
    public const float WoodyardWoodInterval = 3f; // lv1 3s+1
    public const int WoodyardWoodCap = 5; //
    /// <summary> fuelPerMin="4+fireLv*2" = /60 </summary>
    public static float FuelDrainPerSecond(int fireLevel) => (4f + fireLevel * 2f) / 60f;
    // — = §4.2 ① → = —
    public static readonly float[] LightRadiusByLevel = { 0f, 0.55f, 0.95f }; // [lv1][lv2]
    /// <summary> 0= 1= . ×2 </summary>
    public static readonly string[] DispatchNames = { " . ", " . " };
    public static readonly int[] DispatchYield = { 6, 12 }; // ×2
    public static readonly float[] DispatchFrostRisk = { 0f, 0.15f }; // frostRisk=0.05* 15%
    public const float OutsideWarmupFuel = 5f; // —
    // — V1.2 ② + " " —
    public static readonly string[] DigStages =
    {
        " — ",
        " — ",
        " — ",
        " — ",
    };
};
// — §4.1 + D1-D2 —
public const int StartDay = 1;
public const string ColdWaveTeaser = " —D1-D2 D3 ";
public const int MilestoneDay = 7; // 7
public const int OfflineCoins = 240; // M1 8h
public const int StartCitizens = 1; // →2
}
}
// TutorialFlow.cs
using System;
namespace Sy.P05.Newbie
{
    /// <summary>
    /// evt_newbie_step 1-5 ① + ② ③ ④ + ⑤ +
    /// FurnaceSim / / —
    /// Play headless NewbieFlowCheck
    /// </summary>
    public sealed class TutorialFlow
    {
        public int Step { get; private set; }
        readonly Action<string> _hint;
        readonly Action<int> _onStep;
        public TutorialFlow(FurnaceSim sim, Action<string> hint, Action<int> onStep)
        {
            _hint = hint ?? delegate { };
            _onStep = onStep ?? delegate { };
            sim.PhaseChanged += OnPhaseChanged;
            sim.Claimed += () => _hint(" 8h . — 1 . +30 ");
        }
    }
}

```

```

sim.FirewoodAdded += OnFirewoodAdded;
sim.WoodyardBuilt += OnWoodyardBuilt;
sim.FireLevelUp += OnFireLevelUp;
sim.DigStarted += () => _hint(" ..... 10 . ");
sim.DigStageReached += OnDigStageReached;
sim.SurvivorDug += OnSurvivorDug;
sim.DispatchStarted += OnDispatchStarted;
sim.FrostRiskWarned += () => _hint(" . 15% frostRisk — 5 ×2");
sim.OrderDone += OnOrderDone;
sim.Settled += OnSettled;
_hint(" — ");
}
void Advance(int step)
{
    if (Step >= step) return;
    Step = step;
    _onStep(step);
}
void OnPhaseChanged(FurnaceSim.SimPhase phase)
{
    switch (phase)
    {
        case FurnaceSim.SimPhase.AwaitClaim:
            _hint(" 8 — ");
            break;
        case FurnaceSim.SimPhase.AwaitFirewood:
            _hint(" — +30");
            break;
        case FurnaceSim.SimPhase.AwaitBuild:
            _hint(" — lv1 . . lv2 ");
            break;
        case FurnaceSim.SimPhase.AwaitDig:
            _hint(" — ");
            break;
        case FurnaceSim.SimPhase.AwaitDispatch:
            _hint(" +1 ×2");
            break;
        case FurnaceSim.SimPhase.Reveal:
            _hint(" — " + NewbieConfig.ColdWaveTeaser + " ");
            break;
    }
}
void OnFirewoodAdded()
{
    Advance(1);
    _hint(" — + ");
}
void OnWoodyardBuilt()
{
    Advance(2);
}

```

```

        _hint(" lv1          3s+1 · 5 — ");
    }
    void OnFireLevelUp()
    {
        _hint(" lv2 — ③ → ");
    }
    void OnDigStageReached(int stage)
    {
        _hint(NewbieConfig.DigStages[stage - 1] + " " + stage * 25 + "% ");
    }
    void OnSurvivorDug()
    {
        Advance(3);
        _hint(" \ " \"— ·4-24h ");
    }
    void OnDispatchStarted(int node)
    {
        Advance(4);
        _hint(" — " + (node == 1 ? " " : " ") + " ");
    }
    void OnOrderDone(int yield, int node)
    {
        _hint(" +" + yield + (node == 1 ? " ×2· 5 " : " ") + "— ");
    }
    void OnSettled()
    {
        Advance(5);
        _hint(" " + NewbieConfig.StartDay + " → " + (NewbieConfig.StartDay + 1)
            + " — " + NewbieConfig.MilestoneDay + " ");
    }
}

// CampBuilder.cs
using System;
using UnityEngine;
namespace Sy.P05.Night
{
    /// <summary>
    /// $ · C# / 8*lv^1.5+ 5*lv^1.2*$
    /// 0-3 ResourceBank
    /// / / /
    /// Night — FurnaceSim =
    /// </summary>
    public sealed class CampBuilder
    {
        const string LevelsKey = "p05_camp_levels";
        const string WorkersKey = "p05_camp_workers";
        const string DayKey = "p05_camp_day";
        public int[] Levels { get; } = new int[NightConfig.Buildings.Length];
        public int[] Workers { get; } = new int[NightConfig.Buildings.Length];
    }
}

```

```
public int Day { get; private set; } = 1;
public static CampBuilder Load()
{
    var c = new CampBuilder();
    string lv = PlayerPrefs.GetString(LevelsKey, "");
    if (lv.Length > 0)
    {
        var parts = lv.Split(',');
        for (int i = 0; i < parts.Length && i < c.Levels.Length; i++)
            if (int.TryParse(parts[i], out int v)) c.Levels[i] = v;
    }
    string wk = PlayerPrefs.GetString(WorkersKey, "");
    if (wk.Length > 0)
    {
        var parts = wk.Split(',');
        for (int i = 0; i < parts.Length && i < c.Workers.Length; i++)
            if (int.TryParse(parts[i], out int v)) c.Workers[i] = v;
    }
    c.Day = PlayerPrefs.GetInt(DayKey, 1);
    return c;
}
void Save()
{
    var lv = new System.Text.StringBuilder();
    var wk = new System.Text.StringBuilder();
    for (int i = 0; i < Levels.Length; i++)
    {
        if (i > 0) { lv.Append(','); wk.Append(','); }
        lv.Append(Levels[i]);
        wk.Append(Workers[i]);
    }
    PlayerPrefs.SetString(LevelsKey, lv.ToString());
    PlayerPrefs.SetString(WorkersKey, wk.ToString());
    PlayerPrefs.SetInt(DayKey, Day);
    PlayerPrefs.Save();
}
public void AdvanceDay() { Day++; Save(); }
public int Level(int buildingId) => Levels[buildingId - 1];
public bool IsBuilt(int buildingId) => Level(buildingId) > 0;
/// <summary>    =    +1    ·$    </summary>
public int WorkerSlots => Level(2);
/// <summary>    0-3·    false=    </summary>
public bool AssignWorker(int buildingId, int count)
{
    if (buildingId < 1 || buildingId > Levels.Length) return false;
    if (count < 0 || count > 3) return false;
    int used = 0;
    for (int i = 0; i < Workers.Length; i++)
        if (i != buildingId - 1) used += Workers[i];
    if (used + count > WorkerSlots) return false;
}
```

```

        Workers[buildingId - 1] = count;
        Save();
        return true;
    }
    /// <summary>    $    ·lv0→ 1 =1    false </summary>
    public bool TryBuildOrUpgrade(int buildingId, ResourceBank bank)
    {
        var def = Def(buildingId);
        if (def == null) return false;
        if (def.UnlockDay > Day) return false;
        int current = Level(buildingId);
        if (current >= 30) return false;
        int next = current + 1;
        int iron = NightConfig.BuildCostIron(next);
        int grain = NightConfig.BuildCostGrain(next);
        if (!bank.TrySpend(0f, grain, iron)) return false;
        Levels[buildingId - 1] = next;
        Save();
        return true;
    }
    public static NightConfig.BuildingDef Def(int buildingId)
    {
        foreach (var b in NightConfig.Buildings)
            if (b.Id == buildingId) return b;
        return null;
    }
    // ——    $    ·    /    ——
    public (float wood, float grain, float iron) ProductionPerMinute(float heroGatherMul)
    {
        float mul = heroGatherMul * HeroShrineBonus;
        float wood = NightConfig.WoodPerMin(Level(1), Workers[0]) * mul;
        float grain = NightConfig.GrainPerMin(Level(3), Workers[2]) * mul;
        float iron = NightConfig.IronPerMin(Level(4), Workers[3]) * mul;
        return (wood, grain, iron);
    }
    public int WarehouseLevel => Level(6);
    public int DigQueueSize
    {
        get
        {
            int size = Level(7) > 0 ? 1 + Level(7) / 5 : 0;
            return size;
        }
    }
    }
    /// <summary>    -5%/ · 5 =25%·$    </summary>
    public float WatchTowerReduction
        => System.Math.Min(NightConfig.WatchTowerMaxLevel,
            Level(9)) * NightConfig.WatchTowerReductionPerLevel;
    /// <summary>    +10%/5 ·$    </summary>
    public float HeroShrineBonus => Level(10) > 0 ? 1f + Level(10) / 5 * 0.10f : 1f;

```

```

        public static void ClearForTests()
        {
            PlayerPrefs.DeleteKey(LevelsKey);
            PlayerPrefs.DeleteKey(WorkersKey);
            PlayerPrefs.DeleteKey(DayKey);
        }
    }
}

// ColdWave.cs
using System;
using UnityEngine;
namespace Sy.P05.Night
{
    /// <summary>
    ///     § · C#      → → +
    ///     -15%/ · 2 · 3      +
    ///     ≥48h → + = 30% + /
    ///     = ×2×30 § burnMultiplier×nightMinutes —
    ///     = → Night · FurnaceSim
    /// </summary>
    public sealed class ColdWave
    {
        public enum Phase { Idle, Warning, Night, Settled }
        const string LastTickKey = "p05_coldwave_last_ms";
        const string PenaltyNightsKey = "p05_coldwave_penalty_nights";
        /// <summary> · 1 · 2h—§ subscribeAlarm 2h </summary>
        public const float CycleHours = 22f;
        public const float WarningHours = 2f;
        public const float NightHours = NightConfig.NightMinutes / 60f;
        public Phase CurrentPhase(long nowMs, long lastCwMs)
        {
            long elapsed = nowMs - lastCwMs;
            if (elapsed < 0) return Phase.Idle;
            float hours = elapsed / 3_600_000f;
            float cyclePos = hours % CycleHours;
            if (cyclePos < CycleHours - NightHours - WarningHours) return Phase.Idle;
            if (cyclePos < CycleHours - NightHours) return Phase.Warning;
            if (cyclePos < CycleHours) return Phase.Night;
            return Phase.Idle;
        }
        public int PenaltyNights { get; private set; }
        public static ColdWave Load()
        {
            return new ColdWave
            {
                PenaltyNights = PlayerPrefs.GetInt(PenaltyNightsKey, 0),
            };
        }
        void Save()
        {

```

```

        PlayerPrefs.SetInt(PenaltyNightsKey, PenaltyNights);
        PlayerPrefs.Save();
    }
    /// <summary>    $    =    ×2×30min·    /    </summary>
    public static float NightWoodNeed(int fireLevel, CampBuilder camp, HeroDig heroes)
    {
        float burn = NightConfig.FuelPerMin(Mathf.Max(1, fireLevel)) * NightConfig.NightBurnMultiplier;
        float minutes = NightConfig.NightMinutes;
        float reduction = camp.WatchTowerReduction + heroes.ColdConsumptionReduction();
        reduction = Mathf.Min(0.60f, reduction);    //    60%
        return burn * minutes * (1f - reduction);
    }
    /// <summary>    =    =    → +1    3    -15%    </summary>
    public bool SurviveNight(ResourceBank bank, float need)
    {
        if (bank.Wood >= need)
        {
            bank.TrySpend(need, 0f, 0f);
            return true;
        }
        bank.TrySpend(bank.Wood, 0f, 0f);    //
        if (PenaltyNights < NightConfig.FailPenaltyCapNights)
        {
            PenaltyNights++;
            Save();
        }
        return false;
    }
    /// <summary>    $    ≥48h    +    30%    ·    </summary>
    public bool TryExempt(long absentMs, ResourceBank bank, int fireLevel, CampBuilder camp)
    {
        if (absentMs < NightConfig.ExemptAbsentHours * 3_600_000L) return false;
        float gift = NightConfig.FurnaceCapacity(Mathf.Max(1, fireLevel)) * NightConfig.ExemptReturnWoodPct;
        bank.AddWood(gift);
        return true;
    }
    /// <summary>    0.85^    ·    2    →    0.7225    3    -15%    </summary>
    public float ProductionPenalty()
    {
        int nights = Mathf.Min(PenaltyNights, NightConfig.FailPenaltyCapNights);
        if (nights <= 0) return 1f;
        return 1f - nights * NightConfig.FailPenaltyRate;
    }
    public void ClearPenalty()
    {
        PenaltyNights = 0;
        Save();
    }
    public static void Stamp(long nowMs)
    {

```

```

        PlayerPrefs.SetString(LastTickKey, nowMs.ToString());
        PlayerPrefs.Save();
    }
    public static long LastStamp()
    {
        var raw = PlayerPrefs.GetString(LastTickKey, "");
        return long.TryParse(raw, out long v) ? v : 0;
    }
    public static void ClearForTests()
    {
        PlayerPrefs.DeleteKey(LastTickKey);
        PlayerPrefs.DeleteKey(PenaltyNightsKey);
    }
}

// HeroDig.cs
using System;
using UnityEngine;
namespace Sy.P05.Night
{
    /// <summary>
    ///     $ · C#                4h-24h
    ///     P0-1      &lt; min(60min, 20%× )      frozen→awake
    ///     =    1+ 5 +    +1    12 PlayerPrefs
    /// </summary>
    public sealed class HeroDig
    {
        public enum HeroState { Locked, Digging, Awake }
        const string StateKey = "p05_hero_states";
        const string ProgressKey = "p05_hero_progress";
        public HeroState[] States { get; } = new HeroState[NightConfig.Heroes.Length];
        /// <summary>      — /      </summary>
        public float[] Progress { get; } = new float[NightConfig.Heroes.Length];
        public static HeroDig Load()
        {
            var d = new HeroDig();
            string st = PlayerPrefs.GetString(StateKey, "");
            if (st.Length > 0)
            {
                var parts = st.Split(',');
                for (int i = 0; i < parts.Length && i < d.States.Length; i++)
                    if (int.TryParse(parts[i], out int v)) d.States[i] = (HeroState)v;
            }
            string pg = PlayerPrefs.GetString(ProgressKey, "");
            if (pg.Length > 0)
            {
                var parts = pg.Split(',');
                for (int i = 0; i < parts.Length && i < d.Progress.Length; i++)
                    if (float.TryParse(parts[i], System.Globalization.NumberStyles.Float,
                        System.Globalization.CultureInfo.InvariantCulture, out float v)) d.Progress[i] = v;
            }
        }
    }
}

```

```

    }
    return d;
}

void Save()
{
    var st = new System.Text.StringBuilder();
    var pg = new System.Text.StringBuilder();
    for (int i = 0; i < States.Length; i++)
    {
        if (i > 0) { st.Append(','); pg.Append(','); }
        st.Append((int)States[i]);
        pg.Append(Progress[i].ToString("0.###",
            System.Globalization.CultureInfo.InvariantCulture));
    }
    PlayerPrefs.SetString(StateKey, st.ToString());
    PlayerPrefs.SetString(ProgressKey, pg.ToString());
    PlayerPrefs.Save();
}

public int DiggingCount()
{
    int n = 0;
    for (int i = 0; i < States.Length; i++)
        if (States[i] == HeroState.Digging) n++;
    return n;
}

public static NightConfig.HeroDef Def(int heroId)
{
    foreach (var h in NightConfig.Heroes)
        if (h.Id == heroId) return h;
    return null;
}

/// <summary> ResourceBank · CampBuilder </summary>
public bool StartDig(int heroId, ResourceBank bank, CampBuilder camp)
{
    int at = heroId - 1;
    if (at < 0 || at >= States.Length) return false;
    if (States[at] != HeroState.Locked) return false;
    var def = Def(heroId);
    if (def == null) return false;
    if (camp.DigQueueSize <= DiggingCount()) return false;
    if (!bank.TrySpend(0f, 0f, def.PickCost)) return false;
    States[at] = HeroState.Digging;
    Progress[at] = 0f;
    Save();
    return true;
}

/// <summary> dtHours · tick → </summary>
public void Tick(float dtHours)
{
    if (dtHours <= 0f) return;

```

```

    bool changed = false;
    for (int i = 0; i < States.Length; i++)
    {
        if (States[i] != HeroState.Digging) continue;
        var def = NightConfig.Heroes[i];
        Progress[i] += dtHours;
        if (Progress[i] >= def.DigHours)
        {
            States[i] = HeroState.Awake;
            Progress[i] = def.DigHours;
        }
        changed = true;
    }
    if (changed) Save();
}
/// <summary>P0-1          &lt; min(1h, 20%× ) </summary>
public bool BoostWindowOpen(int heroId)
{
    int at = heroId - 1;
    if (at < 0 || at >= States.Length || States[at] != HeroState.Digging) return false;
    var def = NightConfig.Heroes[at];
    float remaining = def.DigHours - Progress[at];
    return remaining <= NightConfig.DigBoostWindowHours(def.DigHours);
}
/// <summary>          id -1=  </summary>
public int TryBoostComplete(int heroId)
{
    if (!BoostWindowOpen(heroId)) return -1;
    int at = heroId - 1;
    States[at] = HeroState.Awake;
    Progress[at] = NightConfig.Heroes[at].DigHours;
    Save();
    return heroId;
}
public int AwakeCount()
{
    int n = 0;
    for (int i = 0; i < States.Length; i++)
        if (States[i] == HeroState.Awake) n++;
    return n;
}
/// <summary>          +20%· +15%          </summary>
public float GatherMultiplier()
{
    float mul = 1f;
    for (int i = 0; i < States.Length; i++)
    {
        if (States[i] != HeroState.Awake) continue;
        var def = NightConfig.Heroes[i];
        if (def.Id == 1) mul += 0.20f;      //      +20%
    }
}

```

```

        if (def.Id == 4) mul += 0.15f;        //      +15%
    }
    return mul;
}
/// <summary>      -10% </summary>
public float ColdConsumptionReduction()
{
    for (int i = 0; i < States.Length; i++)
    {
        if (States[i] != HeroState.Awake) continue;
        if (NightConfig.Heroes[i].Id == 2) return 0.10f;
    }
    return 0f;
}
/// <summary>      +10% </summary>
public float OfflineBonus()
{
    for (int i = 0; i < States.Length; i++)
    {
        if (States[i] != HeroState.Awake) continue;
        if (NightConfig.Heroes[i].Id == 6) return 0.10f;
    }
    return 0f;
}
public static void ClearForTests()
{
    PlayerPrefs.DeleteKey(StateKey);
    PlayerPrefs.DeleteKey(ProgressKey);
}
}
}
// NightConfig.cs
namespace Sy.P05.Night
{
    /// <summary>
    ///      V1.4      X032      ·CEO      0-20261010-1245
    ///      =05_      .md v0.9 2026-09-16 +      V1.3 schema—
    ///      P0-1      =remaining&lt;min(60min,20%×      )
    /// P1-1      lv1-2
    ///      = 1:1 1      =      8h      30
    /// </summary>
    public static class NightConfig
    {
        // ——— §      fireLv 1-30      · 7      ———
        public const int MaxFireLevel = 30;
        public static float FuelPerMin(int fireLv) => 4f + fireLv * 2f;        // §      /
        public static float FurnaceCapacity(int fireLv) => 240f * fireLv;        // §
        public static int LightRadius(int fireLv) => 1 + fireLv / 3;        // §
        // ——— §      ———
        public const int NightMinutes = 30;        // §      nightMinutes
    }
}

```

```

public const float NightBurnMultiplier = 2.0f;           // $ burnMultiplier
public const float FailPenaltyRate = 0.15f;             // $ production -15%/
public const int FailPenaltyCapNights = 2;              // $ 2 3 -15%
public const float ExemptAbsentHours = 48f;            // $ ≥48h
public const float ExemptReturnWoodPct = 0.30f;         // $ = 30%
public const float SubscribeAlarmPct = 0.25f;          // $ <25%
public const int WatchTowerMaxLevel = 5;               // $
public const float WatchTowerReductionPerLevel = 0.05f; // $ -5% -25%
// —— $ . . ——
public static int BuildCostIron(int lv) => (int)System.Math.Round(8f * Pow15(lv));
public static int BuildCostGrain(int lv) => (int)System.Math.Round(5f * Pow12(lv));
static float Pow15(int lv) { float v = lv; return v * Powf(lv, 0.5f); } // lv^1.5
static float Pow12(int lv) { float v = lv; return v * Powf(lv, 0.2f); } // lv^1.2
static float Powf(float v, float e) => (float)System.Math.Pow(v, e);
// —— $ V1.4 = / / / 4 . LiveOps ——
public sealed class BuildingDef
{
    public int Id;
    public string Name;
    public int UnlockDay;
    public bool FirstWave; // V1.4
    public string Effect; //
}
public static readonly BuildingDef[] Buildings =
{
    new BuildingDef { Id = 1, Name = " ", UnlockDay = 0, FirstWave = true, Effect = " 3× × / " },
    new BuildingDef { Id = 2, Name = " ", UnlockDay = 0, FirstWave = false, Effect = " +1 " },
    new BuildingDef { Id = 3, Name = " ", UnlockDay = 1, FirstWave = true, Effect = " 2× × / " },
    new BuildingDef { Id = 4, Name = " ", UnlockDay = 2, FirstWave = true, Effect = " 1.5× × / " },
    new BuildingDef { Id = 5, Name = " ", UnlockDay = 3, FirstWave = false, Effect = " +2 " },
    new BuildingDef { Id = 6, Name = " ", UnlockDay = 5, FirstWave = false, Effect = " +500× " },
    new BuildingDef { Id = 7, Name = " ", UnlockDay = 7, FirstWave = true, Effect = " 5 +1" },
    new BuildingDef { Id = 8, Name = " ", UnlockDay = 10, FirstWave = false, Effect = " 5 +10%" },
    new BuildingDef { Id = 9, Name = " ", UnlockDay = 15, FirstWave = false, Effect = " -5%/ " },
    new BuildingDef { Id = 10, Name = " ", UnlockDay = 20, FirstWave = false, Effect = " 5 +10%" },
};
// $ 1 =1.0 / 2 =1.5 / 3 =2.0
public static float WorkerBonus(int workers) => workers switch
{
    0 => 0f,
    1 => 1f,
    2 => 1.5f,
    _ => 2f,
};
// $ .lv×
public static float WoodPerMin(int woodyardLv, int workers) => 3f * woodyardLv * WorkerBonus(workers);
public static float GrainPerMin(int farmLv, int workers) => 2f * farmLv * WorkerBonus(workers);
public static float IronPerMin(int mineLv, int workers) => 1.5f * mineLv * WorkerBonus(workers);
public const float StorageBase = 500f; // $
public const float WarehouseCapPerLevel = 500f; // $ +500

```

```

public const int SurvivorRationsPerDay = 50; // $ =50 / /
// —— $ . ·V1.4 / ——
public sealed class HeroDef
{
    public int Id;
    public string Name;
    public float DigHours;
    public int PickCost;
    public bool FirstWave; // V1.4
    public string Function; //
}
public static readonly HeroDef[] Heroes =
{
    new HeroDef { Id = 1, Name = " ", DigHours = 4f, PickCost = 20, FirstWave = true, Function = " +20%
    new HeroDef { Id = 2, Name = " ", DigHours = 8f, PickCost = 30, FirstWave = true, Function = " -1
    new HeroDef { Id = 3, Name = " ", DigHours = 12f, PickCost = 35, FirstWave = false, Function = " +1" },
    new HeroDef { Id = 4, Name = " ", DigHours = 16f, PickCost = 40, FirstWave = false, Function = " +15%·
    new HeroDef { Id = 5, Name = " ", DigHours = 18f, PickCost = 45, FirstWave = false, Function = " +15%"
    new HeroDef { Id = 6, Name = " ", DigHours = 20f, PickCost = 50, FirstWave = false, Function = " +10%"
    new HeroDef { Id = 7, Name = " ", DigHours = 22f, PickCost = 55, FirstWave = false, Function = " +25
    new HeroDef { Id = 8, Name = " ", DigHours = 24f, PickCost = 60, FirstWave = false, Function = " +2 " },
};
/// <summary>$ +P0-1 &lt; min(60 , 20%× ) </summary>
public static float DigBoostWindowHours(float totalHours)
{
    float twentyPct = totalHours * 0.2f;
    float cap = 60f / 60f; // 60 =1
    return System.Math.Min(cap, twentyPct);
}
// —— $ D7/D30/D100· ——
public static readonly (int Day, float WoodProd, float WoodDrain, float GrainProd, float GrainDrain, float IronP
{
    (7, 25920f, 22580f, 12960f, 1300f, 6480f, 2020f),
    (30, 51840f, 55980f, 25920f, 3100f, 17280f, 5150f),
    (100, 93312f, 94380f, 69120f, 7000f, 45360f, 14000f),
};
// —— $ / ——
public const int OfflineCapHours = 8; // $ 8h
public const float AdRescueMinFuel = 240f; // $ fillTo80Pct minFuel lv1-2
public const int AdRescueDailyCap = 2; // $ 2/
}
}
// NightOffline.cs
using System;
using UnityEngine;
namespace Sy.P05.Night
{
    /// <summary>
    /// $ ·8h ·P03 OfflinePay
    /// ≥48h + /

```

```

///          .          +10%
///          ResourceBank/HeroDig/ColdWave .
/// </summary>
public static class NightOffline
{
    public const string SeenKey = "p05_night_seen_ms";
    public const int MinClaimMinutes = 60;    // 1
    public static void Stamp(long nowMs)
    {
        PlayerPrefs.SetString(SeenKey, nowMs.ToString());
        PlayerPrefs.Save();
    }
    public static long LastSeen()
    {
        var raw = PlayerPrefs.GetString(SeenKey, "");
        return long.TryParse(raw, out long v) ? v : 0;
    }
    public sealed class Claim
    {
        public long ElapsedMs { get; }
        public bool Claimable => ElapsedMs >= MinClaimMinutes * 60_000L;
        public bool Consumed { get; private set; }
        Claim(long elapsedMs) { ElapsedMs = elapsedMs; }
        </summary>      →      </summary>
        public static Claim Open(long nowMs)
        {
            long seen = LastSeen();
            if (seen <= 0) return new Claim(0);
            return new Claim(nowMs - seen);
        }
        </summary>      8h .
        </summary>      -1= / </summary>
        public int TryCollect(long nowMs, ResourceBank bank, CampBuilder camp, HeroDig heroes, ColdWave cold)
        {
            if (Consumed || !Claimable) return -1;
            float minutes = ResourceBank.OfflineMinutes(ElapsedMs);
            float hours = minutes / 60f;
            var prod = camp.ProductionPerMinute(heroes.GatherMultiplier());
            float penalty = cold.ProductionPenalty();
            bank.ProductionMultiplier = penalty;
            bank.ApplyOffline((long)(minutes * 60_000L),
                prod.wood, prod.grain, prod.iron, camp.WarehouseLevel);
            heroes.Tick(hours);    //
            cold.TryExempt(ElapsedMs, bank, campLevelFire(camp), camp);
            cold.ClearPenalty();    //
            Stamp(nowMs);
            Consumed = true;
            return (int)minutes;
        }
        static int campLevelFire(CampBuilder camp) => 1;    //          UI          cfg

```

```

    }
    public static void ClearForTests()
    {
        PlayerPrefs.DeleteKey(SeenKey);
    }
}

}

// NightWatchHub.cs
using UnityEngine;
using UnityEngine.UI;
namespace Sy.P05.Night
{
    /// <summary>
    ///     V1.4      UGUI·  ×  ×      ·CEO  0-20261010-1245
    ///     HUD      ·  6 +  10      +      +
    ///     +      Night  —      =
    ///     Editor\NightSetup.cs MenuItem      ·
    /// </summary>
    public sealed class NightWatchHub : MonoBehaviour
    {
        static readonly Color IceBlue = new Color(0.659f, 0.773f, 0.855f); // #A8C5DA
        static readonly Color InkBlue = new Color(0.118f, 0.227f, 0.373f); // #1E3A5F
        static readonly Color WarmOrange = new Color(0.957f, 0.529f, 0.204f); // #F38734
        static readonly Color NightBlack = new Color(0.02f, 0.078f, 0.122f);
        static readonly Color PanelDim = new Color(0.05f, 0.12f, 0.20f, 0.92f);

        Canvas _canvas;
        Sprite _square;
        ResourceBank _bank;
        CampBuilder _camp;
        HeroDig _heroes;
        ColdWave _cold;
        Text _woodText, _grainText, _ironText, _coldText, _toastText;
        RectTransform _heroPanel;
        readonly Text[] _buildingLabels = new Text[NightConfig.Buildings.Length];
        readonly Text[] _heroLabels = new Text[NightConfig.Heroes.Length];
        int _lastWood = -1, _lastGrain = -1, _lastIron = -1;
        string _lastCold = "";
        float _tickAccum;

        void Awake()
        {
            Application.targetFrameRate = 30;
            _canvas = FindObjectOfType<Canvas>();
            _square = MakeSquareSprite();
            _bank = ResourceBank.Load();
            _camp = CampBuilder.Load();
            _heroes = HeroDig.Load();
            _cold = ColdWave.Load();
            _bank.ProductionMultiplier = _cold.ProductionPenalty();
            BuildHud();
            BuildColdLine();
        }
    }
}

```

```
        BuildBuildingGrid();
        BuildHeroPanel();
        BuildToastLine();
    }
    void Update()
    {
        //          dt →
        _tickAccum += Time.unscaledDeltaTime;
        if (_tickAccum < 1f) return;
        float dtMinutes = _tickAccum / 60f;
        _tickAccum = 0f;
        var prod = _camp.ProductionPerMinute(_heroes.GatherMultiplier());
        _bank.Tick(dtMinutes, prod.wood, prod.grain, prod.iron, _camp.WarehouseLevel);
        _heroes.Tick(dtMinutes / 60f);
        SyncHud();
    }
    void SyncHud()
    {
        int wood = (int)_bank.Wood, grain = (int)_bank.Grain, iron = (int)_bank.Iron;
        if (wood != _lastWood) { _lastWood = wood; _woodText.text = " " + wood; }
        if (grain != _lastGrain) { _lastGrain = grain; _grainText.text = " " + grain; }
        if (iron != _lastIron) { _lastIron = iron; _ironText.text = " " + iron; }
        float need = ColdWave.NightWoodNeed(1, _camp, _heroes);
        string cold = " " + (_bank.Wood >= need ? " " : " " + (int)need + " ");
        if (cold != _lastCold) { _lastCold = cold; _coldText.text = cold; }
    }
    void BuildHud()
    {
        var root = _canvas.transform;
        _woodText = MakeText(root, " 0", 46, new Vector2(-330, 880), TextAnchor.UpperLeft, 320, 70);
        _woodText.color = WarmOrange;
        _grainText = MakeText(root, " 0", 46, new Vector2(0, 880), TextAnchor.UpperLeft, 320, 70);
        _grainText.color = IceBlue;
        _ironText = MakeText(root, " 0", 46, new Vector2(330, 880), TextAnchor.UpperLeft, 320, 70);
        _ironText.color = IceBlue;
        //
        MakeButton(" ", new Vector2(0, -800), InkBlue, TryClaimOffline);
    }
    void BuildColdLine()
    {
        _coldText = MakeText(_canvas.transform, "", 42, new Vector2(0, 800), TextAnchor.MiddleCenter, 900, 60);
        _coldText.color = IceBlue;
    }
    void BuildBuildingGrid()
    {
        var root = _canvas.transform;
        for (int i = 0; i < NightConfig.Buildings.Length; i++)
        {
            int id = NightConfig.Buildings[i].Id;
            int row = i / 2, col = i % 2;
```

```

        var btn = MakeButton(NightConfig.Buildings[i].Name + " 0",
            new Vector2(-320 + col * 640, 660 - row * 200), InkBlue, () => TryUpgrade(id));
        _buildingLabels[i] = btn.GetComponentInChildren<Text>();
        _buildingLabels[i].color = NightConfig.Buildings[i].FirstWave ? IceBlue : new Color(0.45f, 0.55f, 0.62f)
    }
    RefreshBuildings();
}

void RefreshBuildings()
{
    for (int i = 0; i < NightConfig.Buildings.Length; i++)
    {
        var def = NightConfig.Buildings[i];
        int lv = _camp.Level(def.Id);
        string state = lv > 0 ? "Lv" + lv : (def.UnlockDay > _camp.Day ? "D" + def.UnlockDay : " ");
        _buildingLabels[i].text = def.Name + " " + state;
    }
}

void TryUpgrade(int id)
{
    if (_camp.TryBuildOrUpgrade(id, _bank))
    {
        RefreshBuildings();
        ShowToast(NightConfig.Buildings[id - 1].Name + " ");
    }
    else
    {
        var def = NightConfig.Buildings[id - 1];
        if (def.UnlockDay > _camp.Day) ShowToast(def.Name + " " + def.UnlockDay + " ");
        else ShowToast(" " + NightConfig.BuildCostIron(_camp.Level(id) + 1)
            + " " + NightConfig.BuildCostGrain(_camp.Level(id) + 1) + " ");
    }
}

void BuildHeroPanel()
{
    MakeButton(" ", new Vector2(0, -660), IceBlue, ToggleHeroPanel);
    _heroPanel = MakeOverlayPanel();
    MakeText(_heroPanel, " · ", 68, new Vector2(0, 800), TextAnchor.MiddleCenter, 900, 80).color = IceBlue
    for (int i = 0; i < NightConfig.Heroes.Length; i++)
    {
        int heroId = NightConfig.Heroes[i].Id;
        int row = i / 2, col = i % 2;
        var card = new GameObject("Hero_" + heroId, typeof(RectTransform), typeof(Image), typeof(Button));
        card.transform.SetParent(_heroPanel, false);
        var rt = (RectTransform)card.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = new Vector2(-330 + col * 660, 560 - row * 240);
        rt.sizeDelta = new Vector2(600, 210);
        var img = card.GetComponent<Image>();
        img.sprite = _square;
        img.color = PanelDim;
    }
}

```

```
        _heroLabels[i] = MakeText(card.transform, "", 42, new Vector2(0, 40), TextAnchor.MiddleCenter, 580, 70);
        _heroLabels[i].color = IceBlue;
        MakeText(card.transform, NightConfig.Heroes[i].Function, 32, new Vector2(0, -55),
            TextAnchor.MiddleCenter, 580, 60).color = new Color(0.6f, 0.72f, 0.8f);
        card.GetComponent<Button>().onClick.AddListener(() => TryDig(heroId));
    }
    var close = MakeButton(" ", new Vector2(0, -800), IceBlue, ToggleHeroPanel);
    close.transform.SetParent(_heroPanel, false);
    _heroPanel.gameObject.SetActive(false);
    RefreshHeroes();
}
void RefreshHeroes()
{
    for (int i = 0; i < NightConfig.Heroes.Length; i++)
    {
        var def = NightConfig.Heroes[i];
        switch (_heroes.States[i])
        {
            case HeroDig.HeroState.Locked:
                _heroLabels[i].text = def.Name + " · " + def.PickCost + " ";
                break;
            case HeroDig.HeroState.Digging:
                _heroLabels[i].text = def.Name + " · " + (int)_heroes.Progress[i] + "/" + def.DigHours + "h";
                break;
            default:
                _heroLabels[i].text = def.Name + " · ";
                _heroLabels[i].color = WarmOrange;
                break;
        }
    }
}
void TryDig(int heroId)
{
    if (_heroes.StartDig(heroId, _bank, _camp))
    {
        ShowToast(" " + Def(heroId).Name);
    }
    else
    {
        var def = Def(heroId);
        if (_camp.DigQueueSize <= _heroes.DiggingCount()) ShowToast(" ");
        else ShowToast(" " + def.PickCost + " ");
    }
    RefreshHeroes();
}
static NightConfig.HeroDef Def(int heroId)
{
    foreach (var h in NightConfig.Heroes)
        if (h.Id == heroId) return h;
    return null;
}
```

```

    }
    void ToggleHeroPanel()
    {
        bool show = !_heroPanel.gameObject.activeSelf;
        if (show) RefreshHeroes();
        _heroPanel.gameObject.SetActive(show);
    }
    void TryClaimOffline()
    {
        long now = System.DateTimeOffset.Now.ToUnixTimeMilliseconds();
        var claim = NightOffline.Claim.Open(now);
        int minutes = claim.TryCollect(now, _bank, _camp, _heroes, _cold);
        if (minutes > 0) ShowToast(" " + (minutes / 60) + " ");
        else if (claim.Claimable) ShowToast(" ");
        else ShowToast(" 1 ");
        SyncHud();
    }
    void BuildToastLine()
    {
        _toastText = MakeText(_canvas.transform, "", 40, new Vector2(0, -880), TextAnchor.MiddleCenter, 1000, 70);
        _toastText.color = WarmOrange;
    }
    void ShowToast(string message)
    {
        _toastText.text = message;
        CancelInvoke(nameof(ClearToast));
        Invoke(nameof(ClearToast), 1.6f);
    }
    void ClearToast() => _toastText.text = "";
    // —— UI UGUI· YAML ——
    Button MakeButton(string text, Vector2 pos, Color color, System.Action onClick)
    {
        var go = new GameObject("Btn_" + text, typeof(RectTransform), typeof(Image), typeof(Button));
        go.transform.SetParent(_canvas.transform, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = new Vector2(580, 120);
        var img = go.GetComponent<Image>();
        img.sprite = _square;
        img.color = color;
        var label = MakeText(go.transform, text, 40, Vector2.zero, TextAnchor.MiddleCenter, 580, 120);
        label.color = color == IceBlue ? InkBlue : IceBlue;
        var btn = go.GetComponent<Button>();
        btn.onClick.AddListener(() => onClick());
        return btn;
    }
    RectTransform MakeOverlayPanel()
    {
        var go = new GameObject("NightPanel", typeof(RectTransform), typeof(Image));

```

```

        go.transform.SetParent(_canvas.transform, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = Vector2.zero;
        rt.anchorMax = Vector2.one;
        rt.offsetMin = rt.offsetMax = Vector2.zero;
        var img = go.GetComponent<Image>();
        img.sprite = _square;
        img.color = NightBlack;
        return rt;
    }

    static Text MakeText(Transform parent, string text, int size, Vector2 pos, TextAnchor anchor, float w, float h)
    {
        var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = new Vector2(w, h);
        var t = go.GetComponent<Text>();
        t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
        t.text = text;
        t.fontSize = size;
        t.alignment = anchor;
        t.color = new Color(0.91f, 0.89f, 0.85f);
        return t;
    }

    static Sprite MakeSquareSprite()
    {
        var tex = new Texture2D(8, 8, TextureFormat.RGBA32, false);
        var px = tex.GetPixels32();
        for (int i = 0; i < px.Length; i++) px[i] = new Color32(255, 255, 255, 255);
        tex.SetPixels32(px);
        tex.Apply(false, false);
        return Sprite.Create(tex, new Rect(0, 0, 8, 8), new Vector2(0.5f, 0.5f), 8f);
    }
}

// ResourceBank.cs
using System;
using UnityEngine;
namespace Sy.P05.Night
{
    /// <summary>
    ///     / / .§     · C#         tick
    ///     12 PlayerPrefs=         =     -15%
    ///     +20%
    /// </summary>
    public sealed class ResourceBank
    {
        public const string SaveKey = "p05_night_resources";
    }
}

```

```
const string WoodKey = "p05_night_wood";
const string GrainKey = "p05_night_grain";
const string IronKey = "p05_night_iron";
public float Wood { get; private set; }
public float Grain { get; private set; }
public float Iron { get; private set; }
/// <summary> / * 1 </summary>
public float ProductionMultiplier { get; set; } = 1f;
public static ResourceBank Load()
{
    var b = new ResourceBank
    {
        Wood = PlayerPrefs.GetFloat(WoodKey, 0f),
        Grain = PlayerPrefs.GetFloat(GrainKey, 0f),
        Iron = PlayerPrefs.GetFloat(IronKey, 0f),
    };
    return b;
}
void Save()
{
    PlayerPrefs.SetFloat(WoodKey, Wood);
    PlayerPrefs.SetFloat(GrainKey, Grain);
    PlayerPrefs.SetFloat(IronKey, Iron);
    PlayerPrefs.Save();
}
public float CapFor(int warehouseLevel)
    => NightConfig.StorageBase + warehouseLevel * NightConfig.WarehouseCapPerLevel;
/// <summary> →tick ×dt </summary>
public void Tick(float dtMinutes, float woodPerMin, float grainPerMin, float ironPerMin, int warehouseLevel)
{
    if (dtMinutes <= 0f) return;
    float cap = CapFor(warehouseLevel);
    Wood = Mathf.Min(cap, Wood + woodPerMin * ProductionMultiplier * dtMinutes);
    Grain = Mathf.Min(cap, Grain + grainPerMin * ProductionMultiplier * dtMinutes);
    Iron = Mathf.Min(cap, Iron + ironPerMin * ProductionMultiplier * dtMinutes);
    Save();
}
/// <summary> / false </summary>
public bool TrySpend(float wood, float grain, float iron)
{
    if (Wood < wood || Grain < grain || Iron < iron) return false;
    Wood -= wood;
    Grain -= grain;
    Iron -= iron;
    Save();
    return true;
}
public void AddWood(float amount)
{
    Wood += amount;
```

```
        Save();
    }
    /// <summary> /      .      </summary>
    public void DevSet(float wood, float grain, float iron)
    {
        Wood = wood;
        Grain = grain;
        Iron = iron;
        Save();
    }
    /// <summary> § 8h . .      </summary>
    public float ApplyOffline(long elapsedMs, float woodPerMin, float grainPerMin, float ironPerMin, int warehouseLe
    {
        float minutes = OfflineMinutes(elapsedMs);
        Tick(minutes, woodPerMin, grainPerMin, ironPerMin, warehouseLevel);
        return minutes;
    }
    public static float OfflineMinutes(long elapsedMs)
    {
        if (elapsedMs < 0) return 0f;
        const long MsPerMinute = 60_000L;
        long minutes = elapsedMs / MsPerMinute;
        const long Cap = NightConfig.OfflineCapHours * 60L;
        if (minutes > Cap) minutes = Cap;
        return minutes;
    }
    public static void ClearForTests()
    {
        PlayerPrefs.DeleteKey(WoodKey);
        PlayerPrefs.DeleteKey(GrainKey);
        PlayerPrefs.DeleteKey(IronKey);
        PlayerPrefs.DeleteKey(SaveKey);
    }
}
}
```